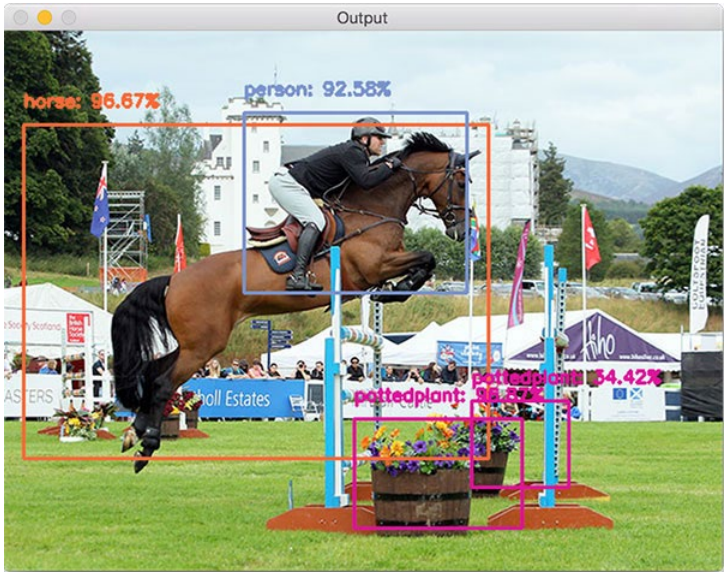
MTRE4820 Machine Learning for Robot Perception

Project #5

Due by 11:59 pm on 10/31/2024 (Wednesday)

In this project, you are required to develop a Python program with Keras implementation of the SSD (Single Shot MultiBox Detector) model **and/or** the Yolo model to detect and track three kinds of target objects (such as paper cups, paper boxes, oranges, balls, and so on) in the live video stream captured from your laptop camera.



In particular, the project requirements are below:

1. It is a group project. In the first line of your Python code, use a comment line to show all group members' names.
2. Please download a Keras implementation of the SSD or Yolo model on github.com.
3. Carefully read the tutorial on the gitgub website and try to understand how to use the SSD or Yolo model to implement fast visual object detection and tracking.
4. Develop your own Python program with the SSD model, which can detect and track the target objects (cups, boxes, oranges, balls, and so on) in the live video stream captured by your laptop camera. You may need to take many pictures and re-train the SSD or Yolo model with them.
5. Your Python program should be able to continuously grab live images from your laptop camera, detect the locations of all target objects in the images, and display the images on screen with the detection results (the bounding boxes), as shown in the image in this page.
6. Each group compresses all your Python code and its related files as "Visual_tracking.zip", and uploads it to the D2L drop box. No project report is required.

Grading Rubric

- 20 points: The Python code and its related files submitted correctly.
- 30 points: The code runs without any syntax errors.
- 30 point: A live window on the computer screen shows the detection and tracking results continuously.
- 20 points. The visual detection results are a kind of success (above 80% accuracy).